

Math 220B HW 2

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1 D

Problem 1

Let R be a semilocal ring and let M and N be two R -modules such that $M \otimes_R N = 0$. Prove or disprove that either $M = 0$ or $N = 0$.

Solution: We'll disprove the statement by providing a counterexample.

Consider the ring $R = \mathbb{Z}/6\mathbb{Z}$, which it is semilocal ring because it only has finitely many maximal ideals (in fact, finitely many ideals since the ring is finite).

Now, consider the canonical projection $\pi_2 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ and $\pi_3 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$, notice that since $6\mathbb{Z} = 2\mathbb{Z} \cap 3\mathbb{Z}$, one has the ideal $6\mathbb{Z} \subset \ker(\pi_2) = 2\mathbb{Z}$ and $6\mathbb{Z} \subset \ker(\pi_3) = 3\mathbb{Z}$. Hence, by Generalized First Isomorphism Theorem, with the canonical projection $\pi_6 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}$, the above two maps uniquely factor into surjective maps $\varphi_2 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2\mathbb{Z}$ and $\varphi_3 : \mathbb{Z}/6\mathbb{Z} \twoheadrightarrow \mathbb{Z}/3\mathbb{Z}$ as follow:

$$\begin{array}{ccc} \mathbb{Z} & \xrightarrow{\pi_2} & \mathbb{Z}/2\mathbb{Z} \\ & \searrow \pi_6 & \nearrow \varphi_2 \\ & \mathbb{Z}/6\mathbb{Z} & \end{array} \quad \begin{array}{ccc} \mathbb{Z} & \xrightarrow{\pi_3} & \mathbb{Z}/3\mathbb{Z} \\ & \searrow \pi_6 & \nearrow \varphi_3 \\ & \mathbb{Z}/6\mathbb{Z} & \end{array}$$

Hence, $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z}$ can be realized as R -module, via the map φ_2 and φ_3 (in particular, one has $\varphi_2(\bar{1}_6) = \bar{1}_2$ and $\varphi_3(\bar{1}_6) = \bar{1}_3$ as the map, based on the above commutative diagrams).

Now, consider the tensor $\mathbb{Z}/2\mathbb{Z} \otimes_R \mathbb{Z}/3\mathbb{Z}$, we claim that it is 0: Recall that $2, 3 \in \mathbb{Z}$ are coprime, with Bezout's Lemma there exists integers $s, t \in \mathbb{Z}$, such that $1 = 2s + 3t$. Then, under the projection map $\pi_6 : \mathbb{Z} \twoheadrightarrow \mathbb{Z}/6\mathbb{Z}$, one knows there exists $\bar{s}_6, \bar{t}_6 \in \mathbb{Z}/6\mathbb{Z}$, such that the following holds:

$$\bar{1}_6 = \bar{2}_6 \cdot \bar{s}_6 + \bar{3}_6 \cdot \bar{t}_6 \tag{1.1}$$

Then, for any $\bar{a}_2 \in \mathbb{Z}/2\mathbb{Z}$ and $\bar{b}_3 \in \mathbb{Z}/3\mathbb{Z}$, one has the following:

$$\begin{aligned} \bar{a}_2 \otimes \bar{b}_3 &= \bar{1}_6 \cdot (\bar{a}_2 \otimes \bar{b}_3) = (\bar{2}_6 \cdot \bar{s}_6 + \bar{3}_6 \cdot \bar{t}_6) \cdot (\bar{a}_2 \otimes \bar{b}_3) \\ &= \bar{s}_6 \cdot ((\bar{2}_6 \cdot \bar{a}_2) \otimes \bar{b}_3) + \bar{t}_6 \cdot (\bar{a}_2 \otimes (\bar{3}_6 \cdot \bar{b}_3)) \\ &= \bar{s}_6 \cdot ((\bar{0}_2 \bar{a}_2) \otimes \bar{b}_3) + \bar{t}_6 \cdot (\bar{a}_2 \otimes (\bar{0}_3 \bar{b}_3)) \\ &= \bar{s}_6 \cdot (\bar{0}_2 \otimes \bar{b}_3) + \bar{t}_6 \cdot (\bar{a}_2 \otimes \bar{0}_3) \\ &= 0 \end{aligned} \tag{1.2}$$

(Note: By definition, one has $\bar{2}_6 \cdot \bar{a}_2 := \varphi_2(\bar{2}_6)\bar{a}_2 = \bar{0}_2\bar{a}_2 = \bar{0}_2$, and same idea for $\mathbb{Z}/3\mathbb{Z}$).

Hence, this proves that $\mathbb{Z}/2\mathbb{Z} \otimes_R \mathbb{Z}/3\mathbb{Z} = 0$. Yet, each $\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/3\mathbb{Z} \neq 0$ as R -module, which is a desired counterexample for the problem.

2 D

Problem 2

Let $0 \rightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \rightarrow 0$ be an exact sequence of R -modules. Assume that M' and M'' are finitely generated R -modules. Prove that M is a finitely generated R -modules. Explain if the converse is true.

Solution: We'll first prove the given statement, then prove the converse is false in general, by providing a counterexample.

I. Proof of Statement:

Given the given sequence is exact, then it implies $M' \xrightarrow{u} M$ is injective, hence $M' \cong \text{im}(u) \subset M$, WLOG we'll denote $\text{im}(u)$ as M' , and $M' \xrightarrow{u} M$ as an inclusion map, and recognize all elements in M' as in M also.

Since M', M'' are finitely-generated, there exists $x_1, \dots, x_n \in M'$ and $y_1, \dots, y_m \in M''$, such that $M' = \sum_{i=1}^n Rx_i$ and $M'' = \sum_{j=1}^m Ry_j$. Then, choose $z_1, \dots, z_m \in M$, such that for each index $j \in \{1, \dots, m\}$, one has $v(z_j) = y_j$. We claim that the collection $\{x_1, \dots, x_n, z_1, \dots, z_m\}$ generates M .

For any $x \in M$, since $v(x) \in M''$, there exists $b_1, \dots, b_m \in R$, such that the following holds:

$$v(x) = \sum_{j=1}^m b_j y_j = \sum_{j=1}^m b_j v(z_j) = v\left(\sum_{j=1}^m b_j z_j\right) \quad (2.1)$$

As a result, one has $v\left(x - \sum_{j=1}^m b_j z_j\right) = 0$, or $x - \sum_{j=1}^m b_j z_j \in \ker(v)$.

Now, using exactness of the sequence, one gets $\ker(v) = \text{im}(u) = M'$, hence there exists $a_1, \dots, a_n \in R$, such that $x - \sum_{j=1}^m b_j z_j = \sum_{i=1}^n a_i x_i$ by the finitely generated property of M' .

As a result, we have $x = \sum_{i=1}^n a_i x_i + \sum_{j=1}^m b_j z_j$, hence all element $x \in M$ can be expressed with some R -linear combination of $\{x_1, \dots, x_n, z_1, \dots, z_m\}$, showing M is finitely generated.

II. Counterexample of the Converse:

Let k be a field, consider the ring $R = k[x_1, x_2, \dots]$, which R is a polynomial ring over k , with (countably) infinite indeterminates. First, let's note that R consists of polynomials that are finite sums of monomials, and each monomial can involve only finite indeterminates.

Consider the ideal $I := (x_1, x_2, \dots)$ (ideal generated by all indeterminates). Then, $I, R, R/I$ are all R -modules, in particular R is a finitely generated R -module (since every $r \in R$ can be written as $r \cdot 1$, so R is generated by 1 as an R -module). Moreover, we naturally have the following exact sequence of R -linear maps:

$$0 \rightarrow I \hookrightarrow R \twoheadrightarrow R/I \rightarrow 0$$

And it's simply because the image of the inclusion $I \hookrightarrow R$ is I , and the kernel of projection $R \twoheadrightarrow R/I$ is also I .

Now, we have a short exact sequence, such that the middle R -module is finitely generated. Yet, we claim that I is not a finitely generated R -module:

Suppose the contrary that I is a finitely generated R -module, then there exists $f_1, \dots, f_n \in I$, such that $I = \sum_{i=1}^n Rf_i = (f_1, \dots, f_n)$ as ideal. Based on our definition, there are only finitely many indeterminates involved in $f_1, \dots, f_n$, say $S = \{x_{i_1}, \dots, x_{i_k}\}$ (and say it's ordered such that $i_1 < \dots < i_k$).

Then, consider the element $x_{i_{k+1}} \in I$: By our assumption, there exists polynomials $g_1, \dots, g_n \in R$, such that $x_{i_{k+1}} = \sum_{i=1}^n g_i f_i$. Which, let $T = \{x_{j_1}, \dots, x_{j_l}\}$ be the (finite) indeterminates involved in the polynomials $g_1, \dots, g_n$. Which, if consider the set of indeterminates $S \cup T \cup \{x_{i_{k+1}}\} = \{x_{i_{k+1}}, x_{m_1}, \dots, x_{m_p}\}$ after reordering, one can realize the elements $x_{i_{k+1}}, f_1, \dots, f_n, g_1, \dots, g_n$ as elements in $k[x_{i_{k+1}}, x_{m_1}, \dots, x_{m_p}] \hookrightarrow k[x_1, x_2, \dots]$, and inside this smaller subring, the equality $x_{i_{k+1}} = \sum_{i=1}^n g_i f_i$ still holds.

Now, consider the evaluation map $k[x_{i_{k+1}}, x_{m_1}, \dots, x_{m_p}] \rightarrow k$, by $x_{i_{k+1}} \mapsto 1$, and all $x_{m_r} \mapsto 0$: Since we've chosen $x_{i_{k+1}}$, such that it's distinct from all x_{i_j} ($1 \leq j \leq k$), then one has x_{i_j} being some of the x_{m_r} , hence $x_{i_j} \mapsto 0$. Also, notice that since each $f_i \in I$, then it has no constant term (since one has $f_i = h_1 x_1 + \dots$ for $h_i \in R$, such that finitely many $h_i \neq 0$; then, all the constant terms in h_i are multiplied with x_i , which is no longer with constant term). Hence, with each $x_{i_j} \mapsto 0$, $f_i \mapsto 0$ also (since it's a polynomial involving only $x_{i_1}, \dots, x_{i_k}$, and with no constant term).

As a result, one has the following:

$$x_{i_{k+1}} = \sum_{i=1}^n g_i f_i \mapsto 0 \tag{2.2}$$

because each $f_i \mapsto 0$. However, this contradicts our assumption that $x_{i_{k+1}} \mapsto 1$. Hence, our assumption must be false, the ideal I cannot be a finitely generated R -module.

Hence, we found an exact sequence of R -modules $0 \rightarrow I \hookrightarrow R \twoheadrightarrow R/I \rightarrow 0$, such that the middle is finitely generated R -module, while the two sides are not guaranteed to be finitely generated (in particular, the left side is not guaranteed). This proves that the converse is in general false.

3 D

Problem 3

Let M and N be flat R -modules, where R is a commutative ring. Prove or disprove that $M \otimes_R N$ is flat R -module.

Solution: We'll prove that $M \otimes_R N$ is also a flat R -module.

Since the tensor functor $(_) \otimes_R (M \otimes_R N) : R\text{-Mod} \rightarrow R\text{-Mod}$ is a right exact functor, to show that $M \otimes_R N$ is flat (or the above tensor functor is exact), one only needs to prove it preserves injective R -linear maps.

Recall that for any R -module K, L, P , there is a unique R -linear isomorphism $\varphi_{KLP} : (K \otimes_R L) \otimes_R P \xrightarrow{\sim} K \otimes_R (L \otimes_R P)$ that satisfies $\varphi_{KLP}((k \otimes l) \otimes p) = k \otimes (l \otimes p)$ for all $k \in K, l \in L$, and $p \in P$.

Now, given an injective R -linear map $f : K \hookrightarrow K'$, since M is flat, then the exactness of the functor $(_) \otimes_R M$ guarantees the map $f \otimes \text{id}_M : K \otimes_R M \rightarrow K' \otimes_R M$ is injective (which the map is given by $(f \otimes \text{id}_M)(k \otimes m) = f(k) \otimes m$ for all $k \in K, m \in M$).

Similarly, since N is flat, then the exactness of the functor $(_) \otimes_R N$ guarantees the map $(f \otimes \text{id}_M) \otimes \text{id}_N : (K \otimes_R M) \otimes_R N \rightarrow (K' \otimes_R M) \otimes_R N$ is injective (which the map is given by $((f \otimes \text{id}_M) \otimes \text{id}_N)((k \otimes m) \otimes n) = (f(k) \otimes m) \otimes n$ for all $k \in K, m \in M$, and $n \in N$).

Finally, consider the following diagram:

$$\begin{array}{ccc} K \otimes_R (M \otimes_R N) & \xrightarrow{f \otimes \text{id}_{M \otimes_R N}} & K' \otimes_R (M \otimes_R N) \\ \varphi_{KMN}^{-1} \downarrow & & \uparrow \varphi_{K'MN} \\ (K \otimes_R M) \otimes_R N & \xrightarrow{(f \otimes \text{id}_M) \otimes \text{id}_N} & (K' \otimes_R M) \otimes_R N \end{array}$$

Notice that the diagram commutes, because for each $k \in K, m \in M$, and $n \in N$, one has the following:

$$(f \otimes \text{id}_{M \otimes_R N})(k \otimes (m \otimes n)) = f(k) \otimes (m \otimes n) \quad (3.1)$$

$$\begin{aligned} & \varphi_{K'MN} \circ ((f \otimes \text{id}_M) \otimes \text{id}_N) \circ \varphi_{KMN}^{-1}(k \otimes (m \otimes n)) \\ &= \varphi_{K'MN} \circ ((f \otimes \text{id}_M) \otimes \text{id}_N)((k \otimes m) \otimes n) \\ &= \varphi_{K'MN}((f(k) \otimes m) \otimes n) \\ &= f(k) \otimes (m \otimes n) \end{aligned} \quad (3.2)$$

Then, since $K \otimes_R (M \otimes_R N)$ is generated by all elements of the form $k \otimes (m \otimes n)$, then the above two calculation shows that the two maps are the same, hence the diagram commutes.

As a result, since $\varphi_{K'MN}$, $(f \otimes \text{id}_M) \otimes \text{id}_N$, and φ_{KMN}^{-1} are all injective, then their composition – which results in $f \otimes \text{id}_{M \otimes_R N}$ – is injective. Therefore, the functor $(_) \otimes_R (M \otimes_R N)$ preserves injective maps, hence is exact. This shows that $M \otimes_R N$ is a flat module.

4 D

Problem 4

Let $\varphi : R \rightarrow S$ be a ring homomorphism and let M be a flat R -module. Prove or disprove that $M \otimes_R S$ is a flat S -module.

Solution: We'll prove that given M is a flat R -module, then $M \otimes_R S$ is a flat S -module.

Since $(M \otimes_R S) \otimes_S (-) : \mathbf{S}\text{-Mod} \rightarrow \mathbf{S}\text{-Mod}$ is right exact, one just needs to prove it preserves injective maps.

For this, let's prove an exercise in Atiyah Macdonald Chapter 2:

Exercise: 2.15

Let A, B be commutative rings, let M be an A -module, P a B -module and N an (A, B) -bimodule (that is, N is simultaneously an A -module and B -module and the two structures are compatible in the sense that $a(xb) = (ax)b$ for all $a \in A, b \in B, x \in N$).

Then, $M \otimes_A N$ is naturally a B -module, $N \otimes_B P$ an A -module, and we have:

$$(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P) \quad (4.1)$$

Proof: We'll prove the three claims in sequence:

I. B -Module Structure on $M \otimes_A N$:

For any $b \in B, m \in M$ and $n \in N$, define the action $B \times (M \otimes_A N) \rightarrow M \otimes_A N$ as $b \cdot (m \otimes_A n) := m \otimes_A (nb)$ on each generator $m \otimes_A n$ (where nb is given by the B -module structure of N), we also define its action on any sum $\sum_{i=1}^k m_i \otimes_A n_i$ by sum of its action on individual $m_i \otimes_A n_i$. As a set map this is well-defined, let's check it does satisfy the B -module properties (which suffices to check for the generators of $M \otimes_A N$). Given any $b, b' \in B$, and any element $m \otimes_A n, m' \otimes_A n' \in M \otimes_A N$, we have the following:

$$\begin{aligned} 1. \quad (b + b') \cdot (m \otimes_A n) &= m \otimes_A (n(b + b')) = m \otimes_A (nb + nb') \\ &= m \otimes_A (nb) + m \otimes_A (nb') \end{aligned} \quad (4.2)$$

$$\begin{aligned} 2. \quad b \cdot (m \otimes_A n + m' \otimes_A n') &= b \cdot (m \otimes_A n) + b \cdot (m' \otimes_A n') \end{aligned} \quad (4.3)$$

(Note: This is by our definition, such that the action on the sum, is the sum of each individual generator after the action).

$$\begin{aligned} 3. \quad (bb') \cdot (m \otimes_A n) &= m \otimes_A (n(bb')) = m \otimes_A (n(b'b)) \\ &= m \otimes_A ((nb')b) = b \cdot (m \otimes_A (nb')) \\ &= b \cdot (b' \cdot (m \otimes_A n)) \end{aligned} \quad (4.4)$$

$$4. \quad 1_B \cdot (m \otimes_A n) = m \otimes_A (n1_B) = m \otimes_A n \quad (4.5)$$

These are all the requirements for being a B -module, hence $M \otimes_A N$ has a B -module structure (by its action on N).

As a side note, this in fact makes $M \otimes_A N$ an (A, B) -bimodule structure, since for any $m \otimes_A n \in M \otimes_A N, a \in A$, and $b \in B$, we have the following:

$$\begin{aligned} a \cdot (b \cdot (m \otimes_A n)) &= a \cdot (m \otimes_A (nb)) = m \otimes_A (a(nb)) \\ &= m \otimes_A ((an)b) = b \cdot (m \otimes_A (an)) \\ &= b \cdot (a \cdot (m \otimes_A n)) \end{aligned} \quad (4.6)$$

Which, the A -action and B -action commutes, showing the bimodule property.

II. A -Module Structure on $N \otimes_B P$:

Similarly, define $A \times (N \otimes_B P) \rightarrow N \otimes_B P$ by $a \cdot (n \otimes_B p) := (an) \otimes_B p$. Based on similar reasonings in **I**, it satisfies all the conditions for A -module, hence $N \otimes_B P$ now can be realized as A -module.

Again, similar to the proof in **I**, the computation shows that $N \otimes_B P$ is an (A, B) -bimodule.

III. Isomorphism:

We aim to show that $(M \otimes_A N) \otimes_B P$ and $M \otimes_A (N \otimes_B P)$ are isomorphic as both A and B -module. For this, we'll construct a B -linear map $(M \otimes_A N) \otimes_B P \rightarrow M \otimes_A (N \otimes_B P)$, and another A -linear map $(M \otimes_A N) \otimes_B P \leftarrow M \otimes_A (N \otimes_B P)$, and prove that these two maps are isomorphism for abelian groups, while both maps are A -linear and B -linear, which will complete the isomorphism.

1. B -linear map:

First, fix arbitrary $p \in P$, and define a map $F_p : M \times N \rightarrow M \otimes_A (N \otimes_B P)$ by $(m, n) \mapsto m \otimes_A (n \otimes_B p)$ for all $(m, n) \in M \times N$. Note that this is A -bilinear, since for all $a, a' \in A$, $m, m' \in M$, and $n, n' \in N$, we have the following:

$$\begin{aligned} F_p(am + a'm', n) &= (am + a'm') \otimes_A (n \otimes_B p) \\ &= a \cdot (m \otimes_A (n \otimes_B p)) + a' \cdot (m' \otimes_A (n \otimes_B p)) \\ &= a \cdot F_p(m, n) + a' \cdot F_p(m', n) \end{aligned} \quad (4.7)$$

$$\begin{aligned} F_p(m, an + a'n') &= m \otimes_A ((an + a'n') \otimes_B p) \\ &= m \otimes_A (a \cdot (n \otimes_B p) + a' \cdot (n' \otimes_B p)) \\ &= a \cdot (m \otimes_A (n \otimes_B p)) + a' \cdot (m \otimes_A (n' \otimes_B p)) \\ &= a \cdot F_p(m, n) + a' \cdot F_p(m, n') \end{aligned} \quad (4.8)$$

Hence, this uniquely factors through the tensor product, as $\overline{F}_p : M \otimes_A N \rightarrow M \otimes_A (N \otimes_B P)$ by $\overline{F}_p(m \otimes_A n) = m \otimes_A (n \otimes_B p)$.

Now, let's define a map $G : (M \otimes_A N) \times P \rightarrow M \otimes_A (N \otimes_B P)$ as this:

$$G\left(\sum_{i=1}^k m_i \otimes_A n_i, p\right) := \overline{F}_p\left(\sum_{i=1}^k m_i \otimes_A n_i\right) \quad (4.9)$$

We claim that G is B -bilinear (which suffices to check each individual tensors). Given any $b, b' \in B$, $(m \otimes_A n), (m' \otimes_A n') \in M \otimes_A N$, and $p, p' \in P$, we have the following:

$$\begin{aligned}
G(b \cdot (m \otimes_A n) + b' \cdot (m' \otimes_A n'), p) &= \overline{F}_p(m \otimes_A (nb) + m' \otimes_A (n'b')) \\
&= m \otimes_A ((nb) \otimes_B p) + m' \otimes_A ((n'b') \otimes_B p) \\
&= m \otimes_A ((n \otimes_B p)b) + m' \otimes_A ((n' \otimes_B p)b') \\
&= b \cdot (m \otimes_A (n \otimes_B p)) + b' \cdot (m' \otimes_A (n' \otimes_B p)) \quad (4.10) \\
&= b \cdot \overline{F}_p(m \otimes_A n) + b' \cdot \overline{F}_p(m' \otimes_A n') \\
&= b \cdot G(m \otimes_A n, p) + b' \cdot G(m' \otimes_A n', p)
\end{aligned}$$

$$\begin{aligned}
G(m \otimes_A n, bp + b'p') &= F_{bp+b'p'}(m \otimes_A n) \\
&= m \otimes_A (n \otimes_B (bp + b'p')) \\
&= m \otimes_A (b \cdot (n \otimes_B p) + b' \cdot (n \otimes_B p')) \\
&= b \cdot (m \otimes_A (n \otimes_B p)) + b' \cdot (m \otimes_A (n \otimes_B p')) \quad (4.11) \\
&= b \cdot \overline{F}_p(m \otimes_A n) + b' \cdot \overline{F}_{p'}(m \otimes_A n) \\
&= b \cdot G(m \otimes_A n, p) + b' \cdot G(m \otimes_A n, p')
\end{aligned}$$

Since G is B -bilinear, there exists a unique B -linear map $\overline{G} : (M \otimes_A N) \otimes_B P \rightarrow M \otimes_A (N \otimes_B P)$ by $\overline{G}((m \otimes_A n) \otimes_B p) = m \otimes_A (n \otimes_B p)$.

2. A -linear map:

Similarly, fix arbitrary $m \in M$, and define a map $H_m : N \times P \rightarrow (M \otimes_A N) \otimes_B P$ by $(n, p) \mapsto (m \otimes_A n) \otimes_B p$ for all $(n, p) \in N \times P$. Using the same logic in the previous part, one can prove that H_m is B -bilinear, hence inducing a unique B -linear map $\overline{H}_m : N \otimes_B P \rightarrow (M \otimes_A N) \otimes_B P$ by $\overline{H}_m(n \otimes_B p) = m \otimes_A (n \otimes_B p)$.

Then, one defines another map $L : M \times (N \otimes_B P) \rightarrow (M \otimes_A N) \otimes_B P$ as follow:

$$L\left(m, \sum_{i=1}^k n_i \otimes_B p_i\right) := \overline{H}_m\left(\sum_{i=1}^k n_i \otimes_B p_i\right) \quad (4.12)$$

Similar to the previous part, we in fact have L being A -bilinear after running through the computation. Hence, this induces a unique A -linear map $\overline{L} : M \otimes_A (N \otimes_B P) \rightarrow (M \otimes_A N) \otimes_B P$ by $\overline{L}(m \otimes_A (n \otimes_B p)) = (m \otimes_A n) \otimes_B p$.

3. Isomorphism:

Given the B -linear map $\overline{G} : (M \otimes_A N) \otimes_B P \rightarrow M \otimes_A (N \otimes_B P)$, and the A -linear map $\overline{L} : M \otimes_A (N \otimes_B P) \rightarrow (M \otimes_A N) \otimes_B P$, if we first look at the two modules' abelian group structure, one have this for all $m \in M, n \in N, p \in P$:

$$\begin{aligned}
\overline{L} \circ \overline{G}((m \otimes_A n) \otimes_B p) &= \overline{L}(m \otimes_A (n \otimes_B p)) = (m \otimes_A n) \otimes_B p \\
\overline{G} \circ \overline{L}(m \otimes_A (n \otimes_B p)) &= \overline{G}((m \otimes_A n) \otimes_B p) = m \otimes_A (n \otimes_B p)
\end{aligned} \quad (4.13)$$

Hence, as homomorphism of abelian groups, $\overline{G}, \overline{L}$ are mutual inverses (since the compositions act as identity on all the generators of the two spaces, in the right composition order).

Now, it suffices to check that $\overline{G}, \overline{L}$ are in fact both A -linear and B -linear.

- For $\overline{G}$, we've constructed it as a B -linear map, so it suffices to prove the A -linear structure (in particular, suffices to prove it preserves the A -action, since additivity is given by the fact that $\overline{G}$ is a homomorphism of abelian group). Given any $a \in A$, and $(m \otimes_A n) \otimes_B p \in (M \otimes_A N) \otimes_B P$, one has the following:

$$\begin{aligned}
\overline{G}(a \cdot ((m \otimes_A n) \otimes_B p)) &= \overline{G}((a \cdot (m \otimes_A n)) \otimes_B p) \\
&= \overline{G}((am) \otimes_A n \otimes_B p) \\
&= (am) \otimes_A (n \otimes_B p) \\
&= a \cdot (m \otimes_A (n \otimes_B p)) \\
&= a \cdot \overline{G}((m \otimes_A n) \otimes_B p)
\end{aligned} \tag{4.14}$$

(Note: Here we use the fact that $(M \otimes_A N) \otimes_B P$ has an A -module structure by letting A act on $M \otimes_A N$, which is proven in **I**, because M, N can be arbitrary).

Hence, $\overline{G}$ preserves A -action, which is A -linear.

- Similarly, for $\overline{L}$, it's constructed as an A -linear map, so it suffices to prove it preserves the B -action. Given any $b \in B$, and $m \otimes_A (n \otimes_B p) \in M \otimes_A (N \otimes_B P)$, one has the following:

$$\begin{aligned}
\overline{L}(b \cdot (m \otimes_A (n \otimes_B p))) &= \overline{L}(m \otimes_A (b \cdot (n \otimes_B p))) \\
&= \overline{L}(m \otimes_A (n \otimes_B (bp))) \\
&= (m \otimes_A n) \otimes_B (bp) \\
&= b \cdot ((m \otimes_A n) \otimes_B p) \\
&= b \cdot \overline{L}(m \otimes_A (n \otimes_B p))
\end{aligned} \tag{4.15}$$

(Note: Again, we use the fact that $M \otimes_A (N \otimes_B P)$ has B -module structure by letting B act on $N \otimes_B P$, proven in **II**).

Hence, $\overline{L}$ preserves B -action, which is a B -linear map.

This realizes $\overline{G}, \overline{L}$ as mutual inverses as both A -linear and B -linear map, hence proving that $(M \otimes_A N) \otimes_B P \cong M \otimes_A (N \otimes_B P)$, as (A, B) -bimodule. In particular, an isomorphism (as both A and B -module) between them that's constructed above, has the following form:

$$\begin{aligned}
\varphi_{MNP} : (M \otimes_A N) \otimes_B P &\xrightarrow{\sim} M \otimes_A (N \otimes_B P) \\
\varphi_{MNP}((m \otimes_A n) \otimes_B p) &= m \otimes_A (n \otimes_B p), \quad \forall m \in M, n \in N, p \in P
\end{aligned} \tag{4.16}$$

□

Since S is an (R, S) -bimodule (where it's both an R -module and S -module, while any $r \in R, s, t \in S$ has $(r \cdot t)s = (\varphi(r)t)s = \varphi(r)(ts) = r \cdot (ts)$, where the dot $\cdot$ represents the R -action on S), then one has the following isomorphism, for any S -module P :

$$\varphi_{MSP} : (M \otimes_R S) \otimes_S P \xrightarrow{\sim} M \otimes_R (S \otimes_S P), \quad \varphi_{MSP}((m \otimes_R s) \otimes_S p) = m \otimes_R (s \otimes_S p) \tag{4.17}$$

Now, given any injective S -linear map $f : P \hookrightarrow P'$ between two S -modules, one simply has $(\text{id}_S) \otimes_S f : S \otimes_S P \hookrightarrow S \otimes_S P'$ be injective (since tensor an S -module over S is isomorphic to the original module).

However, $(\text{id}_S) \otimes_S f$ is also an injective R -linear map, since it's additive, and for all $r \in R, s \in S$, and $p \in P$, one has the following:

$$\begin{aligned}
((\text{id}_S) \otimes_S f)(r \cdot (s \otimes_B p)) &= ((\text{id}_S) \otimes_S f)((r \cdot s) \otimes_B p) \\
&= (r \cdot s) \otimes_B f(p) \\
&= r \cdot (s \otimes_B f(p)) \\
&= r \cdot ((\text{id}_S) \otimes_S f)(s \otimes_B p)
\end{aligned} \tag{4.18}$$

Hence, with M being a flat R -module, the R -linear map $\text{id}_M \otimes_R ((\text{id}_S) \otimes_S f) : M \otimes_R (S \otimes_S P) \hookrightarrow M \otimes_R (S \otimes_S P')$ is also injective.

Similar to the previous one, we also have $\text{id}_M \otimes_R ((\text{id}_S) \otimes_S f)$ being an injective S -linear map, since it's additive, for any $m \in M$, $s, t \in S$, and $p \in P$, one has the following:

$$\begin{aligned}
(\text{id}_M \otimes_R ((\text{id}_S) \otimes_S f))(s \cdot (m \otimes_R (t \otimes_S p))) &= (\text{id}_M \otimes_R ((\text{id}_S) \otimes_S f))(m \otimes_R (s \cdot (t \otimes_S p))) \\
&= m \otimes_R (((\text{id}_S) \otimes_S f)((st) \otimes_S p)) \\
&= m \otimes_R ((st) \otimes_S f(p)) \\
&= m \otimes_R (s \cdot (t \otimes_S f(p))) \\
&= s \cdot (m \otimes_R (t \otimes_S f(p))) \\
&= s \cdot [(\text{id}_M \otimes_R ((\text{id}_S) \otimes_S f))(m \otimes_R (t \otimes_S p))]
\end{aligned} \tag{4.19}$$

For simplicity, let's define $\psi := \text{id}_M \otimes_R ((\text{id}_S) \otimes_S f)$ as the injective S -linear map.

Finally, one has the following diagram:

$$\begin{array}{ccc}
(M \otimes_R S) \otimes_S P & \xrightarrow{(\text{id}_M \otimes_R S) \otimes_S f} & (M \otimes_R S) \otimes_S P' \\
\varphi_{MSP} \downarrow & & \uparrow \varphi_{MSP'}^{-1} \\
M \otimes_R (S \otimes_S P) & \xrightarrow{\psi} & M \otimes_R (S \otimes_S P')
\end{array}$$

Notice that this big diagram commutes, because for all $m \in M$, $s \in S$, and $p \in P$, one has the following:

$$((\text{id}_M \otimes_R S) \otimes_S f)((m \otimes_R s) \otimes_S p) = (m \otimes_R s) \otimes_S f(p) \tag{4.20}$$

$$\begin{aligned}
\varphi_{MSP'}^{-1} \circ \psi \circ \varphi_{MSP}((m \otimes_R s) \otimes_S p) &= \varphi_{MSP'}^{-1} \circ \psi(m \otimes_R (s \otimes_S p)) \\
&= \varphi_{MSP'}^{-1}(m \otimes_R (s \otimes_S f(p))) \\
&= (m \otimes_R s) \otimes_S f(p)
\end{aligned} \tag{4.21}$$

This shows that $\varphi_{MSP'}^{-1} \circ \psi \circ \varphi_{MSP} = (\text{id}_M \otimes_R S) \otimes_S f$ as S -linear maps. Hence, with each map on the LRS being injective, the RHS is also injective.

This shows that $(M \otimes_R S) \otimes_S (_)$ is an exact functor (as it preserves injectivity), hence $M \otimes_R S$ can be realized as a flat S -module.

5 D

Problem 5

Let $\varphi : R \rightarrow S$ be a ring homomorphism such that M is projective as an R -module. Prove or disprove that $M \otimes_R S$ is projective as S -module.

Solution: We'll prove that if M is projective as R -module, then $M \otimes_R S$ is projective as S -module.

Since M is projective as an R -module, there exists another R -module N , such that $M \oplus N \cong \bigoplus_{i \in I} Re_i$ a free R -module (where each $Re_i \cong R$ as R -module). Then, since tensor product „commutes“ with direct sum (up to isomorphism), hence one has the following isomorphism as R -modules:

$$(M \otimes_R S) \oplus (N \otimes_R S) \cong (M \oplus N) \otimes_R S \cong \left(\bigoplus_{i \in I} Re_i \right) \otimes_R S \cong \bigoplus_{i \in I} (Re_i \otimes_R S) \quad (5.1)$$

As a remark, all these isomorphism respects the S -module structure: Given any family of R -modules $\{M_i\}_{i \in \Lambda}$, the R -linear isomorphism $\varphi : \left(\bigoplus_{i \in \Lambda} M_i \right) \otimes_R S \xrightarrow{\sim} \bigoplus_{i \in \Lambda} (M_i \otimes_R S)$ is given as:

$$\varphi\left(\left((m_i)_{i \in \Lambda}\right) \otimes s\right) = (m_i \otimes s)_{i \in \Lambda} \quad (5.2)$$

Which, this map is also S -linear, as any $(m_i)_{i \in \Lambda} \in \bigoplus_{i \in \Lambda} M_i$, and $s, t \in S$, satisfy the following:

$$\begin{aligned} \varphi\left(s \cdot \left((m_i)_{i \in \Lambda} \otimes t\right)\right) &= \varphi\left((m_i)_{i \in \Lambda} \otimes (ts)\right) = (m_i \otimes (ts))_{i \in \Lambda} \\ &= (s \cdot (m_i \otimes t))_{i \in \Lambda} = s \cdot (m_i \otimes t)_{i \in \Lambda} \end{aligned} \quad (5.3)$$

Hence, φ is also an S -linear isomorphism. Apply this logic to the 1st and 3rd isomorphism relation, they're isomorphic as S -module (the middle part is true because S is preserved).

Now, look at each $Re_i \cong R$, one has each $Re_i \otimes_R S \cong S$ as R -module (and a specific R -linear isomorphism is given by $\varphi : R \otimes_R S \xrightarrow{\sim} S$, $\varphi(r \otimes s) = r \cdot s = \varphi(r)s$).

However, notice that this φ is also an S -linear map, because it's additive, and also every $r \in R$ and $s, t \in S$ satisfy the following:

$$\varphi(s \cdot (r \otimes t)) = \varphi(r \otimes (ts)) = r \cdot (ts) = \varphi(r)(ts) = (\varphi(r)t)s = s \cdot (r \cdot t) = s \cdot \varphi(r \otimes t) \quad (5.4)$$

(Note: Here the S -module structure on $R \otimes_R S$ is by letting S act on S).

Hence, it's not only an R -linear isomorphism, but in fact an S -linear isomorphism. As a consequence, with the S -module structure on $R \otimes_R S$ being isomorphic to S , one has:

$$(M \otimes_R S) \oplus (N \otimes_R S) \cong \bigoplus_{i \in I} (Re_i \otimes_R S) \cong \bigoplus_{i \in I} S \quad (5.5)$$

Where the final part is a free S -module. This realizes $M \otimes_R S$ as a direct summand of a free S -module, hence it's projective as S -module.

6 ND

Problem 6

Let $\varphi : R \rightarrow S$ be a ring homomorphism and let M be an injective R -module such that $M \otimes_R S \neq 0$. Prove or disprove that $M \otimes_R S$ is injective S -module.

Solution: Guess: $R = \mathbb{Z}$, $S = \mathbb{Z}_{(p)}$ for some prime p , choose $M = \mathbb{Q}$, then $M \otimes_R S \neq 0$ I believe (even though it's fxxking hard to check). I think $M \otimes_R S \cong M$ as S -module in this case (bc the description of $\mathbb{Z}_{(p)}$ can be realized as a subring of $\mathbb{Q}$, and one can simply move all coefficients into $\mathbb{Q}$ by divisibility of $\mathbb{Q}$).

Is $\mathbb{Q}$ an injective S -mod? Yes, bc the fraction field of S is $\mathbb{Q}$, and S is a PID.

Maybe it's true then...

7 HD (Type it up)

Problem 7

Let $\varphi : R \rightarrow S$ be a ring homomorphism such that S is flat R -algebra. Prove or disprove that φ is injective.

Solution: We'll disprove it by providing a counterexample.

Consider the ring $R = \mathbb{Z}/6\mathbb{Z}$, recall that it has a projection $\varphi : R \rightarrow \mathbb{Z}/3\mathbb{Z}$ by $\varphi(\bar{1}_6) = \bar{1}_3$, this realizes $\mathbb{Z}/3\mathbb{Z}$ as an R -algebra, while the map φ is not injective (since $\varphi(\bar{3}_6) = \bar{3}_3 = 0$).

(Basically, the idea is that as R -modules, $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z}$ because the section $\mathbb{Z}/3\mathbb{Z} \rightarrow \mathbb{Z}/6\mathbb{Z}$ by $\bar{1}_3 \mapsto \bar{2}_6$ splits, as a result $S = \mathbb{Z}/3\mathbb{Z}$ is a projective R -mod, hence flat)..

8 HD (Type it up)

Problem 8

Let $\varphi : R \rightarrow S$ be an injective ring homomorphism such that S is a field. Prove or disprove that S is a flat R -algebra.

Solution: Notice that R is a subring of a field, hence an integral domain. Given K as fraction field of R , one realizes S as a field extension of K , hence $S \cong \bigoplus_{i \in I} K$ as K -vector space. Notice that they're also isomorphic as R -modules (simply because a K -linear map is also an R -linear map, by restricting the scalars). So, it suffices to argue that K is flat or not (since direct sum commutes with tensor).

K is injective: Consider any inclusion of ideals $I \hookrightarrow R$, w claim that $I \otimes_R K \rightarrow R \otimes_R K \cong K$ is injective:

- First, its image are characterized by finite sum of $a_i \cdot \frac{p_i}{q_i}$, where $a_i \in I$ and $\frac{p_i}{q_i} \in K$ (or, $p_i, q_i \in R$). As a result, one has $a_i \cdot \frac{p_i}{q_i} \in IK$ (the finite sum of products of elements in I and elements in K).
- Now, notice that we have an inverse: Every element in IK can be written as $\frac{a_i}{q_i}$ where $a_i \in I$ and $q_i \in R$, hence define the inverse by $\frac{a_i}{q_i} \mapsto a_i \otimes \frac{1}{q_i}$. This is well-defined as multiplying

by $r \in R$ has: $r \frac{a_i}{q_i} \mapsto a_i \otimes \frac{r}{q_i} = r \left(a_i \otimes \frac{1}{q_i} \right)$. On the other hand, it can be checked that this is a mutual inverse.

So, $I \hookrightarrow R$ has $I \otimes_R K \hookrightarrow R \otimes_R K$, showing K is flat.

If R is a field then we're done (everything is flat).

If R is not a field, how to characterize flatness? We know K definitely can't be free, and there are cases where it's not projective (EX: $\mathbb{Q}$ over $\mathbb{Z}_{(p)}$ covered in class).

9 ND

Problem 9

Let R be a dedekind domain and let $I \subset R$ be an ideal. Prove or disprove that I is a flat R -module.

Solution:

10 ND

Problem 10

Let R be a PID and let M be a finitely generated flat R -module. Prove or disprove that M is a free R -module.

Solution: Known information: All submodules of a finitely generated free module over R (a PID) is finitely generated and free.

Which, the quotient is of the form $R^n \oplus \frac{R}{I_1} \oplus \dots \oplus \frac{R}{I_t}$. So, it suffices to prove that if M is of such form, it must be free (or prove that some R/I over a PID can be flat).

However, R/I I don't think is flat in general...

11 HD (Type it up)

Problem 11

Let R be a PID and let M be an R -module. Prove that M is a submodule of an injective R -module.

Solution: Consider K as the fraction field of R , then any $\bigoplus_{i \in I} K$ is injective (because it's divisible coordinate wise!)

For any M an R -module, there exists free R -module $\bigoplus_{i \in J} R$ that surjects onto M , hence $M \cong (\bigoplus_{i \in J} R)/L$ for some submodule L of the free module.

Now, consider the inclusion of R -modules $L \hookrightarrow \bigoplus_{i \in J} R \hookrightarrow \bigoplus_{i \in J} K$, and the quotient $\bigoplus_{i \in J} K \twoheadrightarrow (\bigoplus_{i \in J} K)/L$. Then, the map $\bigoplus_{i \in J} R \rightarrow (\bigoplus_{i \in J} K)/L$ has kernel precisely L , showing $(\bigoplus_{i \in J} R)/L \hookrightarrow (\bigoplus_{i \in J} K)/L$ by its natural identification.

So, M injects into $(\bigoplus_{i \in J} K)/L$, which is divisible (since it's a module theoretic quotient of $\bigoplus_{i \in J} K$ which is divisible).

12 HD (Type it up)

Problem 12

Let R be a commutative ring and let m be a given maximal ideal of R . Suppose M is an R -module such that $M_m = 0$. Prove or disprove that $M = 0$.

Solution: I guess we can take $R = \mathbb{Z}$, $m = 2\mathbb{Z}$, and $M = \mathbb{Z}/3\mathbb{Z}$. Then, M_m denotes all the elements of the form $\frac{\bar{a}_3}{d}$, where $d \notin 2\mathbb{Z}$ (i.e. we can divide by 3!) So in particular, one has the following:

$$3 \cdot (1 \cdot \bar{a}_3 - 1 \cdot \bar{0}_3) = 3 \cdot \bar{a}_3 = \bar{0}_3 \in M \quad (12.1)$$

This shows that $\frac{\bar{a}_3}{1} = \frac{0}{1}$ in M_m , showing $M_m = 0$, while $M \neq 0$.
(But if swapping to all maximal ideal m I think it's true).

13 HD (Type it up)

Problem 13

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be two ring homomorphisms such that S is R -flat and T is S -flat. Prove or disprove that T is R -flat.

Solution: Consider the following isomorphism as (R, S) -bimodules:

$$(R \otimes_R S) \otimes_S T \cong R \otimes_R (S \otimes_S T) \cong R \otimes_R T \cong T \quad (13.1)$$

As a result, if $I \hookrightarrow R$ is inclusion of ideal, one has the injection $I \otimes_R S \hookrightarrow R \otimes_R S \cong S$ as R -module (which is also an injective S -linear map, bc S is an (R, S) -bimodule). This is by R -flatness of S .

As a result, one has the following injection:

$$(I \otimes_R S) \otimes_S T \hookrightarrow (R \otimes_R S) \otimes_S T \cong S \otimes_S T \cong T \quad (13.2)$$

Which, using the bimodule isomorphism proven in problem 2, we have:

$$I \otimes_R T \cong I \otimes_R (S \otimes_S T) \hookrightarrow R \otimes_R (S \otimes_S T) \cong R \otimes_R T \quad (13.3)$$

Hence, T is also R -flat.

14 HD (Type it up)

Problem 14

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms such that T is R -flat. Prove or disprove that φ can not be surjective.

Solution: Consider the isomorphism $\mathbb{Z}/30\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/3\mathbb{Z} \cong \mathbb{Z}/5\mathbb{Z}$, I think this is also a $\mathbb{Z}/30\mathbb{Z}$ -mod. Then, take successive quotient we're good.

15 HD (Type it up)

Problem 15

Let $R \xrightarrow{\varphi} S \xrightarrow{\psi} T$ be ring homomorphisms such that T is R -flat. Prove or disprove that ψ can not be surjective.

Solution: Look at 14 for counterexample, they apply to each other.

16 ND

Problem 16

Let $R \xrightarrow{\varphi} S$ be ring homomorphism which is flat. Let M be an R -module such that $M \otimes_R S$ is torsion-free S -module. Is M torsion-free R -module?

Solution: The counterexample I can thought of are all not flat. I want to say it's true.